CSE400: Fundamentals of Probability in Computer science

Section 1

Computer Networks

Group 16

Optimizing UAV Deployment for Multi-UAV Cooperative Sensing

Background and Motivation

Course Perspective

- Application of probability in real-world systems
- Use of:
  - Random variables
  - Gaussian noise
  - Hypothesis testing
- Introduction to Monte Carlo simulation & optimization

Real-World Perspective

- UAVs used in:
  - Surveillance
  - Defense systems
  - Disaster management
- Real environments are:
  - Noisy
  - Uncertain

Base Paper Overview

Main Idea

- Use multiple UAVs for cooperative target detection
- Improve detection using:
  - Eigenvalue-based method
  - Optimization techniques

Key Contributions

- Models detection as probabilistic problem
- Uses:
  - Monte Carlo simulation
  - Statistical detection theory
- Applies PSO for optimization

Our Work

- Compared:
  - Deterministic approach
  - Randomized approaches (PSO, GA)

Problem Formulation

- UAV-based target detection system
- Environment contains:
  - Noise
  - Uncertainty
- Hypothesis Testing
  - H0: Noise only
  - H1: Signal + Noise
- Objective
  - Maximize Probability of Detection (Pd)
  - Minimize False Alarm (Pfa)
- Challenges
  - Noisy wireless signals
  - Uncertain UAV positions
  - Signal fluctuations

Mathematical Model

- Binary Hypothesis Testing Model:

$$H_0 : y(k) = w(k)$$
$$H_1 : y(k) = x(k) + w(k)$$
$$w(k) \sim \mathcal{N}(0, \sigma^2)$$

- Covariance matrix is computed

$$R_y = \frac{1}{K} \sum_{k=0}^{K-1} y(k)\, y^H(k)$$

- Gaussian Approximation:

$$T \sim \mathcal{N}\left(1, \frac{2}{K}\right) \quad \text{under } H_0$$
$$T \sim \mathcal{N}\left(1 + \text{SNR}, \frac{2(1+\text{SNR})^2}{K}\right) \quad \text{under } H_1$$

- (Test Statistic) Detection using eigenvalues:
  - $T = \lambda_{max} / \lambda_{min}$

- Thresold:

$$\lambda_{EBD} = 1 + \sqrt{\frac{2}{K}}\, Q^{-1}(P_{fa})$$

- Probability :

$$P_{fa} = P(T \geq \lambda_{EBD} \mid H_0)$$
$$P_d = P(T \geq \lambda_{EBD} \mid H_1)$$

- Cooperative Detection:

$$P_n = 1 - \prod_{m=1}^{M} (1 - P_{m,n}) \qquad P_{sum} = \sum_{n=1}^{N} P_n$$

- Optimization Variables:

$$X_i, Y_i \sim U(0, A)$$
$$\eta_i \sim U(-\pi, \pi)$$

Probability Model

- False Alarm constraint:

$$P_{fa} = P(T \geq \lambda_{EBD} \mid H_0)$$
$$P_d = P(T \geq \lambda_{EBD} \mid H_1)$$
$$P_{fa} \leq \alpha$$

- Inequality Bounds:

$$\text{Markov: } P(T \geq \gamma) \leq \frac{\mathbb{E}[T]}{\gamma}$$
$$\text{Chebyshev: } P(T \geq \gamma) \leq \frac{\sigma^2}{(\gamma - \mu)^2}$$

- Spatial Constraints:

$$0 \leq X \leq 1800$$

Minimum UAV–UAV separation

Minimum UAV–target separation

- Optimization Constraint:

$$\text{Fitness} = P_{sum} - \lambda \cdot \text{Penalty}$$

Deterministic Approach

- UAV placement is fixed:

X = 900 (constant)
Even spacing along Y-axis

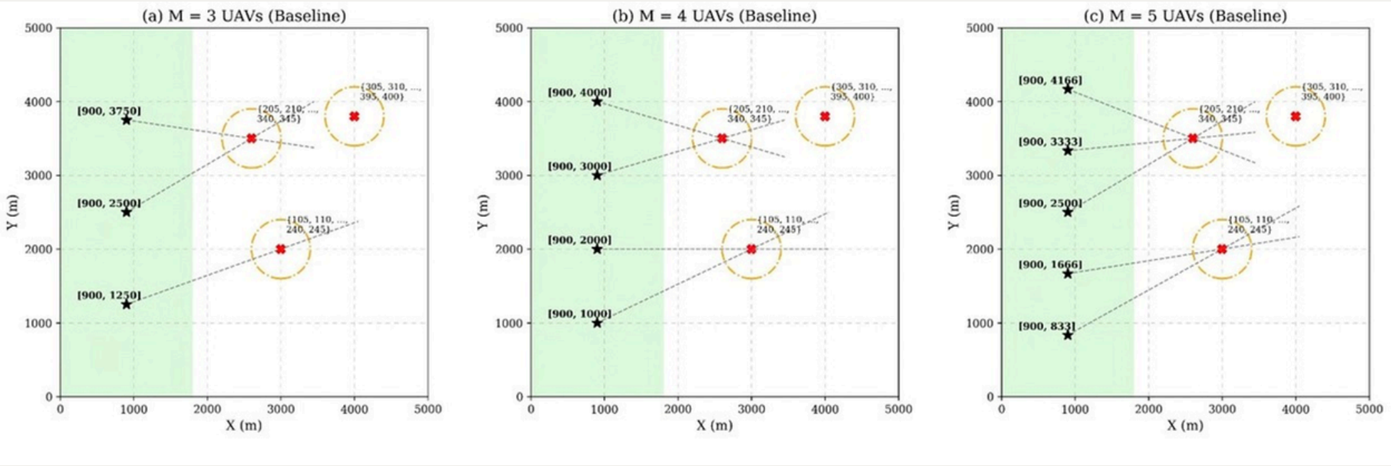
- Evaluate for:

M = 3, 4, 5 UAVs
Although, more efficient if 2 UAVs are used, as it not always optimal

- Features:

Simple and easy to implement
Provides uniform spatial coverage
Serves as a baseline for comparison

Results



Increasing UAV count improves coverage density and reduces average distance to targets, which enhances SNR and detection performance. However, the deterministic placement lacks adaptability, so performance gains are limited.

- Limitations

Fixed placement → no adaptability
Cannot respond to changing target positions
May lead to suboptimal detection performance

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Deterministic Strategy Observations

- More UAVs → better coverage, improves spatial coverage
- Closer UAV → higher detection probability
- Even spacing gives stable performance

But:

- Not optimal for all target positions

Randomized Approach

What approach did we choose?

- We implemented a randomized optimization strategy using:
  - Genetic Algorithm (GA)
  - Particle Swarm Optimization (PSO)
- Instead of fixed UAV placement, we generate and evaluate multiple random configurations

How does it help in our model?

- Randomness is introduced in:
  - UAV position generation
  - Optimization updates (mutation, velocity, etc.)
- Each configuration is evaluated using Monte Carlo simulation:
  - Different noise realizations
  - More realistic performance estimation

Why did we choose this approach?

- The UAV placement problem has a large and complex search space
- Deterministic baseline:
  - Uses fixed positions
  - Cannot adapt to different scenarios
- Randomized algorithms help:
  - Explore many possible UAV placements
  - Avoid poor fixed configurations
  - Improve system performance under uncertainty

Key Impact in Our System

- Converts static system → adaptive system
- Enables better exploration of deployment strategies
- Achieves higher performance than deterministic baseline

Coding & Simulation

Simulation Setup

- Implemented in Python (NumPy, SciPy)
- Multi-UAV deployment in 2D area
- System parameters:
  - Area: 1750 × 5000
  - UAV height: 500
  - False alarm probability (P_FA = 0.001)
- Constraints:
  - Minimum UAV-target distance
  - Minimum UAV-UAV separation

Detection Modeling

- SINR computed using distance & antenna alignment
- Detection probability got using statistical model
- Objective: Maximize P_sum (total detection probability)
- Penalty-based fitness ensures constraint satisfaction

```
AREA_X = 1750
AREA_Y = 5000
H = 500.0

K = 1000
P_FA = 0.001
THETA_DEG = 20.0

NOISE_DBM = -80.0
P_N_DBM = 20.0
```

Implementation Modules

- Signal Modeling
  - compute_sinr_vectorized()
  - compute_psum()
- Optimization
  - baseline_placement()
  - run_pso()
  - run_ga()
- Evaluation
  - fitness_evaluation()

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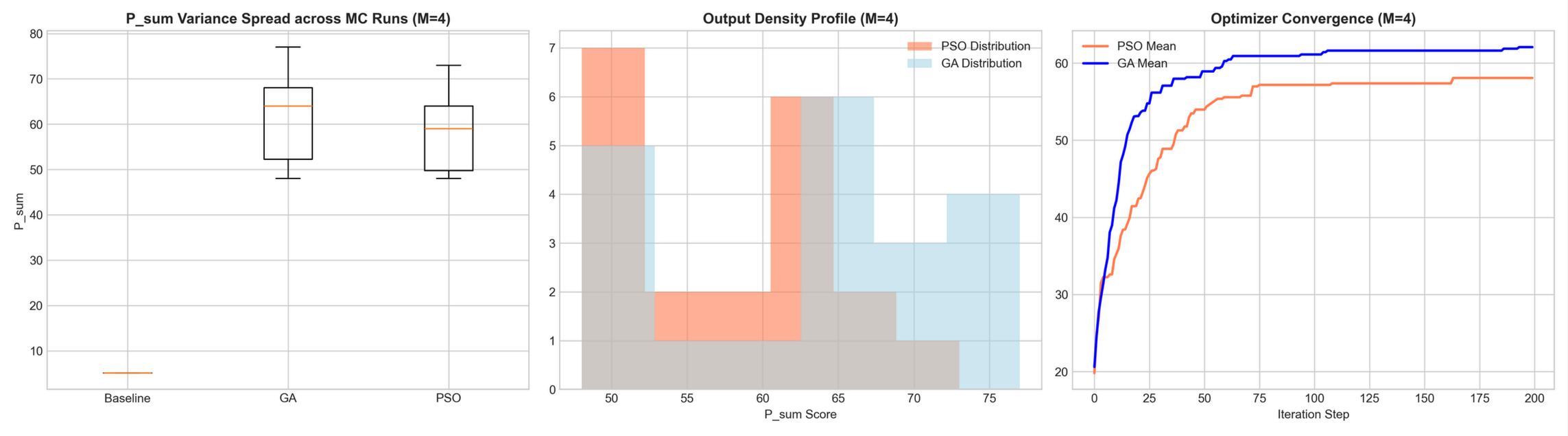
Simulation Results & Analysis



- Baseline shows very poor performance (~5)
- GA & PSO achieve significantly higher P_sum (~60+)
- GA shows slightly higher median performance than PSO

- GA distribution shifted towards higher values
- PSO shows slightly wider spread

- PSO converges faster in early iterations
- GA improves steadily over time

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Deterministic vs Randomized Results

Statistical Validation

- Tested using 20 Monte Carlo runs
- Target positions slightly changed in each run
- Performance measured using P_sum’

Analysis

- **Deterministic (Baseline):** fixed approach, low performance (~5)
- **Randomized (PSO, GA):** explores solutions adaptive, high performance (~60+)
- **PSO:** faster convergence
- **GA:** better final result for more iterations

	A	B	C	D
1	**UAVs**	**Baseline**	**GA**	**PSO**
2	M=3	1.03	57.87	**64.31**
3	M=4	5.07	**62.07**	58.07
4	M=5	8.07	60.47	**62.12**

Inference – Domain Perspective

- Accurate detection is improved via cooperative UAV sensing
- Compared to a single UAV, many UAVs offer better coverage.
- Probabilistic modeling manages uncertainty and noise in the real world.
- Placement at random improves system performance.
- Useful in:
  - Surveillance
  - Defense systems
  - Search & rescue operations

Inference – Computer Science Perspective

- Utilizing probability theory in practical systems
- Monte Carlo simulation used to assess performance
- Putting optimization methods into practice:
  - PSO (fast convergence)
  - GA (better exploration)
- Importance of randomization in algorithm design
- Integration of:
  - Mathematical modeling
  - Coding and simulation

References

1. Research paper:

R. Moliya, D. K. Patel, M. López-Benítez and A. Fakhrcddine, "Optimizing UAV Deployment for Enhanced Detection Performance in Multi-UAV Cooperative Sensing," 2025 National Conference on Communications (NCC), New Delhi, India, 2025, pp. 1-6, doi: 10.1109/NCC63735.2025.10983909. keywords: {Monte Carlo methods;Fluctuations;Navigation;Noise;Directional antennas;Autonomous aerial vehicles;Sensors;Iterative methods;Particle swarm optimization;Vehicle dynamics;Unmanned Aerial Vehicles;Eigenvalue-Based Detection;Particle Swarm Optimization;Cooperative Sensing},

2. Ali Mirjalili. (2018, March 30). Learn Particle Swarm Optimization (PSO) in 20 minutes [Video]. YouTube. https://www.youtube.com/watch?v=JhgDMAm-imI